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Gain Enhancement in Microstrip Patch Antenna With High Negative Refractive Index 3D-Metamaterial Inspired Superstrate for Wireless Applications

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ABSTRACT This article presents a 3D-metamaterial (3D-MTM) structure to enhance microstrip patch antenna gain. A 3D-MTM unit cell exhibits an MNG (μ-negative) property with a high negative refractive index over the 2.36-2.53 GHz frequency range resonating at 2.45 GHz. Therefore, a 4 $\times$ 4 array of the proposed 3D-MTM unit cell is analyzed and used as a superstrate over a patch antenna and a 2 $\times$ 2 array antenna to achieve high gain for wireless applications. The simulated and measured results of the designed antenna show that the gain is enhanced by about 5dBi and 3.2dBi. In addition, the half power beam width reduced in E-plane is 23° and 12.85° for a patch antenna and 2 $\times$ 2 array antenna, respectively.

INDEX TERMS Microstrip antenna, array, 3D-metamaterial superstrate, negative refractive index, gain.

I. INTRODUCTION

The Metamaterial (MTM) development has been explored in various communication subsystems, especially antennas. A low-profile antenna has poor performance in antenna applications, such as low gain, directivity, and efficiency. The literature suggests a number of methods for increasing the gain and directivity of planar antennas, including Artificial Magnetic Conductor (AMC) [1], loading a metasurface [2], electromagnetic Band-Gap (EBG) [3], reflector layer [4], Frequency Selective Surface (FSS) [5]. However, these approaches have limitations, such as exhibiting narrow bandwidth, moderate gain, complex fabrication, requiring complex power distribution, etc. In comparison to a conventional array antenna, a planar antenna with MTM does not require a power distribution system or complicated feed to attain high gain [6]. MTMs are synthetic constructs that alter electromagnetic (EM) wave properties by systematically arranging their unit cells in either a periodic or non-periodic configuration. MTMs can be employed as a substrate or supersubstrate in antennas for enhancing gain and bandwidth [7], [8]. Some of the distinctive characteristics associated with MTM include a zero refractive index and a negative refractive index [9], [10].The MTM slab with zero refractive index causes the transmission angle to be zero Because of this, every surface of a slab with a zero refractive index radiates in a direction that is perpendicular to the face [11]. The directivity and gain of a patch antenna can both be improved when an MTM layer is employed as the superstrate [12], [13]. A miniaturized patch antenna with a significant gain was designed with an Electric-LC (ELC) resonator and Hilbert fractal as substrate and superstrate layer to achieve high permittivity and permeability. The gain is improved by 8.1dBi at 534MHz [14]. A single-layer phi shaped MTM exhibits double negative medium(DNG) and is used as a superstrate with a rectangular patch antenna

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that obtained an enhanced gain of 1.15dBi at 2.45GHZ [15]. In another study, a compact antipodal Vivaldi antenna with V-shaped MTM exhibits epsilon negative metamaterial (ENG). This ENG MTM enhances the reflection coefficients across a broad range from 24 to 30 GHz, with antenna gain ranging from 10.9 to 13.82dBi [16]. A circular spiral-based MTM exhibits positive permittivity and negative permeability. This MTM is used as the patch antenna substrate with an enhanced gain of 2.2dBi [17]. A multiband CPW-fed antenna with a superstrate consisting of several symmetric split-ring resonators(SSRRs) exhibits negative permittivity and permeability. The maximum gain of 6.74dBi is obtained as compared to the antenna without metamaterial [18]. A high-gain microstrip patch antenna with zero index cloved-shaped metamaterial superstrate is investigated to achieve an improved antenna gain of about 6.1dBi at 2.45GHZ [19]. In order to improve gain, directivity, and radiation properties in antenna applications, MTM with a negative refractive index can also be employed as a flat lens to focus electromagnetic waves. The phase variation of electromagnetic waves is extremely tiny or zero, which enhances gain. This is an intriguing property of electromagnetic waves traveling from a negative refractive index superstrate to the air medium [20], [21]. A nested schurig shaped MTM structure as a superstrate exhibits double positive characteristics, and a zero index medium resonates at dual-band 2.45 GHz and 3.5 GHz with an enhanced gain of 2.5 dBi and 3.2 dBi, respectively [22]. A diagonal repeating pattern of MTM unit cells is integrated into the substrate to enhance gain in lower band and upper band. The maximum gain is improved in the lower resonance band around 3.69dBi [23]. A circular patch antenna loaded with a double-sided nested split ring resonator exhibiting positive refractive index resonating at 2.45GHz, resulting in a 5 dBi gain enhancement [24]. A negative index four-layer ELC MTM superstrate achieves a high gain and wideband antenna. The structure resonates at 9.4GHz with an improved gain of 10dBi compared to a conventional patch antenna [25]. For wireless communication, a superstrate layer consists of patches etched uniformly on both sides of the substrate. It exhibits a high negative refractive index in the frequency range of 4 to 8 GHz. The patch has an increased gain and bandwidth of 13dBi and 380MHz at 5.8GHz [27]. A rectangular patch with an I-shaped unit cell superstrate resonates at multiband frequencies and enhances the maximum antenna gain at 6.18GHz, 9.65GHz and 11.45GHz are 4.19dBi, 2.4dBi, and 5.68dBi respectively [26]. Multiple resonance frequencies were achieved using a quintuple circular SRR loaded with a rectangular microstrip patch antenna [28]. This article advances our existing concept [29], Considering antenna with low gain enhancement in [17] and [22], antenna gain increased as the number of superstrate layers increased, and the overall geometry also increased [14], [25]. Therefore, the aim of this study is to present a comprehensive solution for addressing these limitations. A microstrip-based superstrate with a high negative refractive index achieved a high gain.

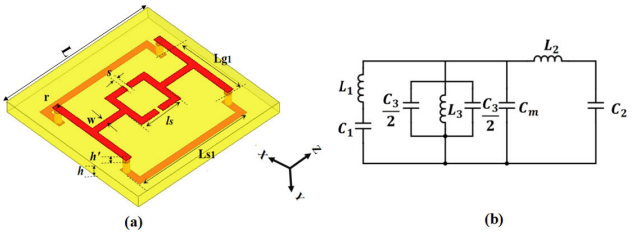


FIGURE 1. 3D-MTM unit cell (a) geometrical configuration (b) equivalent circuit.

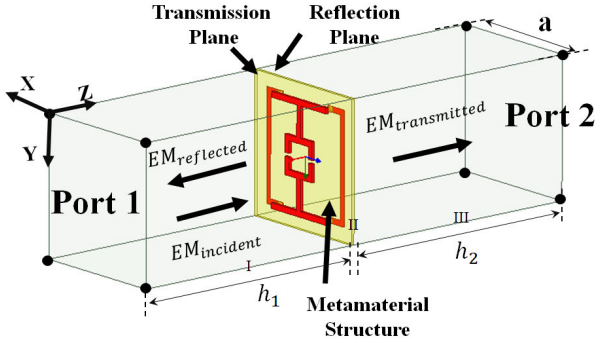


FIGURE 2. Simulation setup of 3D-MTM unit cell.

TABLE 1. Dimensions of proposed 3D-MTM unit cell (in *mm*).

Parameters	L_{s1}	L_{g1}	l_s	w
Values	20	14	7	1
Parameters	s	r	h	h'
Values	0.65	0.5	0.51	0.5

A single slab of 4 × 4 periodic structure of a 3D-MTM cells exhibits a high negative refractive index behavior are shown in previous understanding [30] and used as a superstrate results in enhancement of gain and bandwidth. It is well known that for an EM waves radiated from the patch antenna and free space are coupled and directed towards the direction of the patch antenna. Because of the extremely minimal or zero phase variation of EM waves, gain is increased. A crucial design factor for enhancing antenna gain is the separation between the superstrate and patch antenna. This article is organized as follows, Section II is the configuration and characterization of a negative index 3D-MTM unit cell. Section III describes a 3D-MTM unit cell and array performance setup. Section IV focused on the gain enhancement of patch antenna and 2 × 2 array antenna using high negative refractive index 3D-MTM superstrate.

II. 3D-MTM UNIT CELL: CONFIGURATION AND CHARACTERIZATION

The configuration of the high negative refractive index 3D-MTM unit cell is shown in Figure 1(a). The proposed 3D-MTM unit cell consists of a square bracket structure, dual split ring, metallic vias, and an air gap. It is comprised of two

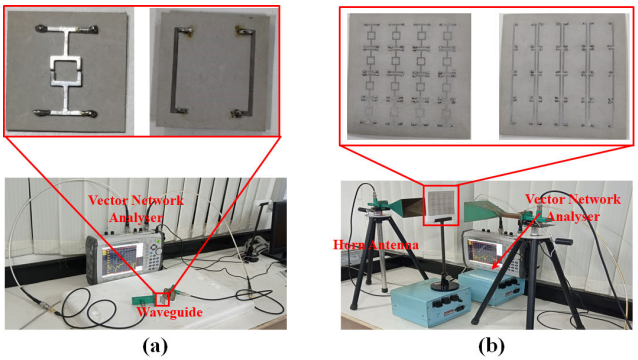


FIGURE 3. Fabricated prototype with measurement setup (a) 3D-MTM unit cell (b) 3D-MTM 4 ×4 array.

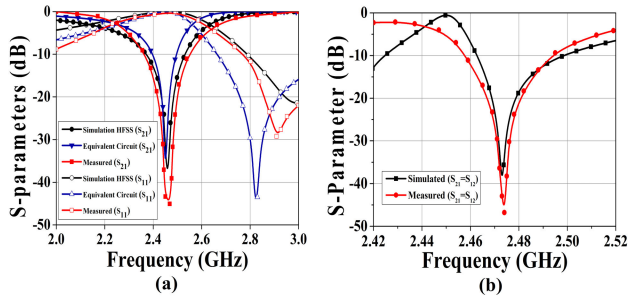


FIGURE 4. Simulated and measured S-parameters of 3D-MTM (a) unit cell (b) 4 × 4 array.

layers: a square bracket structure is printed on the bottom side of the substrate and another dual split ring is placed top above the substrate with an air gap (h'). There is an air gap that acts as a capacitive coupling gap between the substrate and the dual split ring resonator.

The square bracket structure is printed on Rogers 5880 substrate have $\varepsilon_r = 2.2$, $\tan\delta = 0.0009$ and height (h). The length (L) of the substrate is about $21mm$. The square bracket structure and dual split ring are connected with metallic vias of radius (r). The initial dimensions were obtained as λ/10 at 2.45 GHz frequency using the typical sub-wavelength principle [4]. The dimension of the proposed 3D-MTM unit cell is shown in Table 1. The equivalent circuit of the 3D-MTM unit cell is shown in Figure 1(b). The equivalent circuit model can be used to calculate the resonant frequency [29], which is represented as:

$$f = \frac{1}{2\pi\sqrt{L_1C_1L_2C_2}} \tag{1}$$

The effective inductance L_1, L_2 and capacitance C_1, C_2 of the dual split ring and square bracket structure is represented by [25]. For capacitance,

$$C_1(F) = \varepsilon_0\varepsilon_r\frac{A'}{l_s} \tag{2}$$

and

$$C_2(F) = \varepsilon_0\varepsilon_r\frac{A''}{L_{s1}} \tag{3}$$

where, free space permittivity is ε_0, relative permittivity is ε_r. The L_{s1}, l_s, and L_{g1} are the length of square bracket and dual ring, respectively. Similarly, the cross-section area is A', A''.

For inductance,

$$L_1(nH) = 2\times10^{-4}l_s \left[\ln\frac{l_s}{w+t'} + 1.193 + 0.02235\frac{w+t}{l}\right]k_g \tag{4}$$

and

$$L_2(nH) = 2\times10^{-4}L_{s1} \left[\ln\frac{L_{s1}}{w+t} + 1.193 + 0.02235\frac{w+t'}{L_{s1}}\right]k_g \tag{5}$$

where k_g is the correction factor, which is calculated by the equation $k_g = 0.57 - 0.145\ln\left(\frac{L}{h}\right)\varepsilon_r h'$. It depends on three factors such as substrate length (L) and thickness (h), the dielectric constant of the substrate (ε_r), and the distance between the top of the dielectric substrate and dual split ring resonator (h'), respectively. The copper thicknesses of the dual split ring and square bracket structures are $t = 0.508$ mm and $t' = 0.036$ mm, respectively. The series inductance and capacitance L_1,L_2, C_1,C_2 of the dual split ring and square bracket structure is primarily formed LC resonant circuits, significant in achieving the desired resonant frequency. While, the parallel components L_3 and C_3 represent the self-inductance and capacitance of metallic via's, as seen in the equivalent circuit model, and are serially coupled to the inductance L_1 and L_2 of the dual split ring and square bracket structure. There is a coupling gap C_m that exists between the top of the substrate and the dual split ring, which is physically evaluated and expressed as $C_m = \varepsilon_r\frac{d}{h'}$, where ε_r is the permittivity of 3D-MTM substrate, d is the cross-sectional area of the coupling gap, h' is the height of the coupling gap between dual the top of the dielectric substrate and the dual split ring [30]. Hence, L_3, C_3, and C_m do not affect the obtained desired frequency and these elements are not explicitly considered and their impact on calculating the resonant frequency was considered minimal. It influences the coupling, impedance, and other electrical factors that remain crucial and should not be neglected.

$$L_3 = \frac{\mu_0\mu_r}{\pi}\cosh\left(\frac{h+h'}{r}\right)t' \tag{6}$$

and

$$C_3 = \frac{\varepsilon_0\varepsilon_r A}{h+h'} \tag{7}$$

where, μ_0 and μ_r are the free space permeability and substrate permeability. r and $h+h'$ are the radius and height of the metallic via. $A = 2rh$ is the cross-section area of the metallic via. Table 2 shows optimized values of an equivalent circuit which is simulated using ANSYS HFSS circuit simulation tool.

TABLE 2. Extracted equivalent circuit values of proposed 3D-MTM unit cell.

Parameters	L_1 (nH)	L_2 (nH)	L_3 (nH)
Values	0.1776	6.122	3.516
Parameters	C_1 (pF)	C_2 (pF)	C_3 (pF)
Values	5.049	0.8146	0.2146

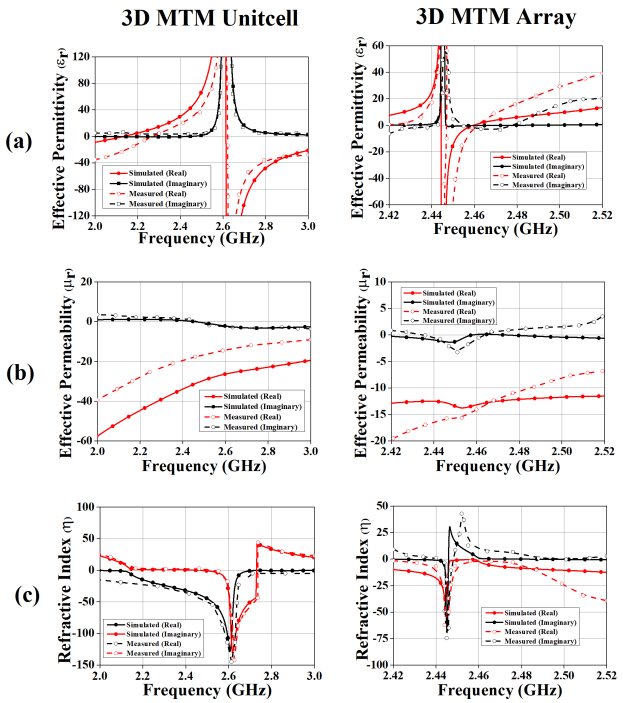


FIGURE 5. Simulation and measurement of effective parameters (a) permittivity (b) permeability (c) refractive index.

III. 3D-MTM UNIT CELL AND ARRAY PERFORMANCE SETUP

The proposed model performance is determined using the Nicolson-Ross-Weir (NRW) Method [14]. Figure 2 shows the simulation setup of the unit cell inside the waveguide. The perfect electric field boundary is applied along the X-axis. The perfect magnetic field boundary is applied along the Y-axis. Two waveguide ports are used along the Z-axis, and EM wave propagation occurs along it. Hence, the electric fields of the transmitted and reflected waves are combined in the transmission and reflection planes, respectively in three different sections E_I, E_{II}, and E_{III}. It is written as:

$$E_I = e^{(-\gamma_0 h_1)} + B_1 e^{\gamma_0 h_1} \tag{8}$$

$$E_{II} = B_2 e^{(-\gamma(h_1+h+h'))} + B_3 e^{\gamma(h_1+h+h')} \tag{9}$$

$$E_{III} = B_4 e^{(-\gamma_0(h+h'+h_2))} \tag{10}$$

where, $\gamma = j[(\omega/c)^2 - (2\pi/\lambda_c)^2]^{1/2}$ and $\gamma_0 = j[(\omega\mu_r\varepsilon_r/c)^2 - (2\pi/\lambda_c)^2]^{1/2}$ are the propagation constant for air and unit cell, respectively. Here, c and ω represent the speed of light and the angular frequency, respectively. The cut-off wavelength and waveguide width are denoted by $\lambda_c = 2a$ and a, respectively. The boundary condition of the electric and magnetic fields can be used to calculate the constant $B_i (i = 1,2,3,4)$. The wave guides overall height is $h_{total} = h_1 + h + h' + h_2$. Here, the receptive port heights are h_1 and h_2, h is substrate height and h' is the air gap height of the unit cell model. Therefore, the scattering parameters [26] of these two ports are calculated as:

$$S_{11} = P_1^2\left[\frac{\Gamma(1-T^2)}{1-\Gamma^2T^2}\right] \tag{11}$$

$$S_{22} = P_2^2\left[\frac{\Gamma(1-T^2)}{1-\Gamma^2T^2}\right] \tag{12}$$

$$S_{12} = S_{21} = P_1P_2\left[\frac{\Gamma(1-T^2)}{1-\Gamma^2T^2}\right] \tag{13}$$

where the two-port reference plane transformations are P_1 and P_2. For the proposed 3D-MTM model, the reflection coefficient and transmission coefficient are denoted by the variables Γ and T. Equations (11) and (13) are used to compute the scattering parameter. Next, the 4×4 array structure is extended from the single unit cell. The dimensions of 4×4 array structure are $100 \times 100\ mm^2$. The distance between the two unit cells is $1.5mm$. Simulations of array performance are performed using an approach similar to that of unit cells, resulting in simulated S-parameters. The 3D-MTM unit cell prototype is shown in Figure 3(a) placed between two waveguide ports for measurement, and its experimental setup validates the proposed design. A free space measurement system [11] is used to validate the array structure. Fabricated prototype and measurement setup for a 3D-MTM array as shown in Figure 3(b). To determine the appropriate reflector in the incident wave, two wide band identical horn antennas were used. First, a copper plate is placed between two horn antennas. Figure 3(b) shows the fabricated prototype of the 3D-MTM 4×4 array and its measurement setup. Two wide band identical horn antennas were used to determine the appropriate reflector in the incident wave. A copper plate is first placed between two horn antennas. Then, the fabricated prototype is mounted between the horn antennas.

Figure 4(a and b) shows the S-parameters of 3D-MTM unit cell and 4×4 array model, respectively. Figure 4(a) shows S-parameter values of the equivalent circuit, simulated and measured for the 3D-MTM unit cell. The magnitude of the calculated S-parameters matches the measurement quite well. During the measurement setup alignment, a tiny difference was observed between the measured and simulated values. It resonates at 2.45 GHz with a band gap of 2.36-2.53 GHz. Figure 4(b) shows the simulated and measured transmission coefficient ($S_{21} = S_{12}$) for the proposed 4×4 3D-MTM array model, it exhibited to resonate at 2.45 GHz frequency but due to the capacitance effect occurred in the array of unit cells increases the frequency and resonant at 2.475 GHz which cover the band gap of 2.36-2.53 GHz. The magnitude of the S-parameters matches the measurement quite well in both cases. A slight variation is observed between the simulated and measured results due to fabrication tolerance,

measurement methodologies, or mutual interaction between unit cell in 4 × 4 array prototypes.

Figure 5(a-c) shows the simulated and measured effective permittivity, effective permeability, and refractive index for the proposed model. The effective medium parameters are calculated as: the complex permittivity

$$\mu_r = \frac{2}{jk_0h}\left[\frac{1-H_2}{1+H_2}\right] \tag{14}$$

the complex permeability

$$\varepsilon_r = \frac{2}{jk_0h}\left[\frac{1-H_1}{1+H_1}\right] \tag{15}$$

and refractive index

$$\eta = \sqrt{\varepsilon_r \mu_r} \tag{16}$$

where $H_1 = (S_{21} + S_{11})$ and $H_2 = (S_{21} - S_{11})$, k_0 the wave number and η is the refractive index. Figure 5(a and b) presents the simulated and measured effective parameters such as (effective permittivity and permeability) in real and imaginary for unit cell and 4 × 4 array. It is extracted from the S-parameters in the frequency range of 2.46-2.53 GHz resonating at 2.45GHz. In the specified frequency range, the real component of permittivity for the unit cell has a high positive range with values between 36 and 42. In the resonance frequency of 2.45 GHz, the simulated and observed values are 40 and 36, respectively. The real part of permittivity for the array case shows the positive value of ≈40 in the resonant frequency. For both unit cell and array cases at the resonant frequency, the imaginary part of permittivity is near zero. The real part of permeability for the unit cell in the desired frequency range are negative value of −40 to −36. The simulated negative value is −35 and the measured negative value is −29 at 2.45 GHz. At the resonant frequency, the negative value is −14 for the array case. The imaginary part of permeability has a positive range value in the desired frequency range for both unit cell and array cases. It is observed that these values indicate the proposed 3D-MTM behaves as a μ-negative MTM. The effective refractive index for the unit cell and array cases are shown in Figure 5(c). For the unit cell, the real part of the refractive index has a high negative value of −26 to −36 in the frequency range of 2.36-2.53 GHz and a negative value of −30 at resonant 2.45 GHz. For the array case, the real part of the refractive index is −15 to −25 in the frequency range of 2.36-2.53GHz. It is observed that the refractive index is found to be a highly negative value of ≈-25. Due to the high permittivity value, the refractive index obtained a negative value. The imaginary part of the refractive index is positive. It is observed that these values clearly indicate the proposed 3D-MTM 4 × 4 array behaves as a μ-negative MTM. In addition, the compactness of the MTM is calculated using the effective medium ratio (EMR) [27] is calculated as:

$$EMR = \frac{\lambda}{L_{s1}} \tag{17}$$

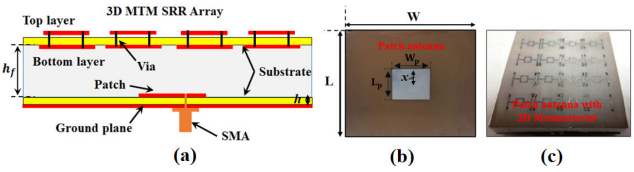


FIGURE 6. Patch antenna (a) side view (b) patch antenna prototype (c) patch antenna with 3D-MTM prototype.

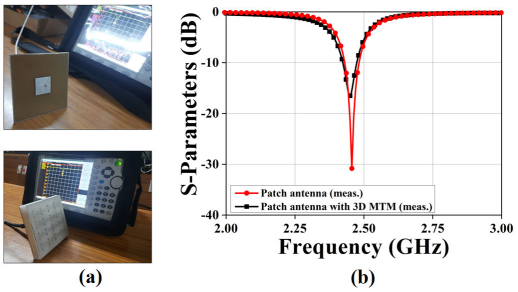


FIGURE 7. Patch antenna (a) measurement setup of patch antenna and with 3D-MTM superstrate (b) measured S-parameter of patch antenna and with 3D-MTM superstrate.

where λ and L_{s1} represents the wavelength and overall length of the proposed 3D-MTM unit cell. The proposed unit cell EMR is 6.16 for the dimension of 20 × 20 × $0.51mm^3$ at the resonant frequency of 2.45 GHz. The unit cell may display negative permittivity and/or permeability in the effective medium area if the EMR is higher than 4, which is the optimal value. This characteristic resulted in the design of 3D-MTM unit cell array being designed in microstrip patch antenna and array antenna to enhance the gain. The next section IV describes the investigation of 4 × 4 3D-MTM as a superstrate with a microstrip patch antenna.

IV. 3D-MTM SUPERSTRATE BASED MICROSTRIP PATCH ANTENNA

A. PATCH ANTENNA WITH 3D-MTM SUPERSTRATE

The 4 × 4 3D-MTM array is used as a superstrate to improve the performance of a patch antenna at the resonant frequency of 2.45 GHz. The side view and fabricated prototype of a patch antenna and with 3D-MTM superstrate is shown in Figure 6. The patch antenna is designed with the basic design equations using [31]. It is printed on a substrate FR-4 with ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$), and a thickness of $1.6mm$. The size of the substrate and ground plane are $0.82\lambda_\circ \times 0.82\lambda_\circ$. The radiating element ($L_p \times W_p$) is $0.23\ \lambda_\circ \times 0.23\lambda_\circ$ is printed on top of the substrate. The feed point (x) is located offset from the center of the patch antenna is about $0.045\lambda_\circ$; fed with 50 Ω impedance matching. The 3D-MTM superstrate is placed above the patch antenna with foam between them. The optimized height (h_f) of the 3D-MTM superstrate is chosen as $13mm$. An effective coupling gap between the superstrate and microstrip patch antenna is essential because it allows electromagnetic waves to be coupled and increases radiation.

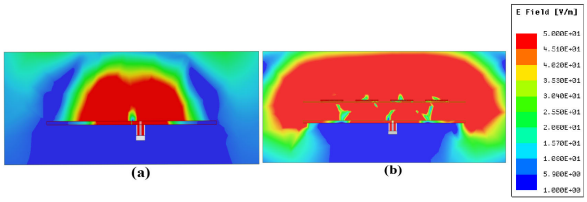


FIGURE 8. Electric field distribution (a) patch antenna (b) patch antenna with 3D-MTM superstrate.

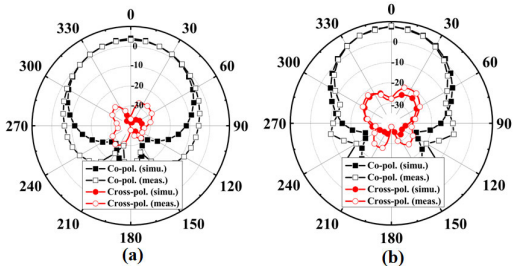


FIGURE 9. Simulated and measured E-plane radiation pattern (a) patch antenna (b) patch antenna with 3D-MTM superstrate.

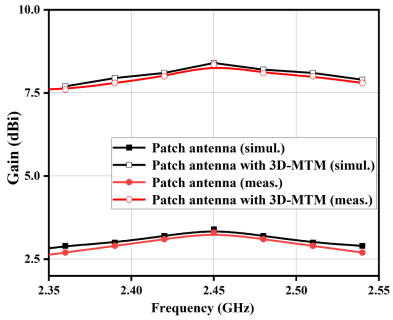


FIGURE 10. Gain comparison of patch antenna without and patch antenna with 3D-MTM superstrate.

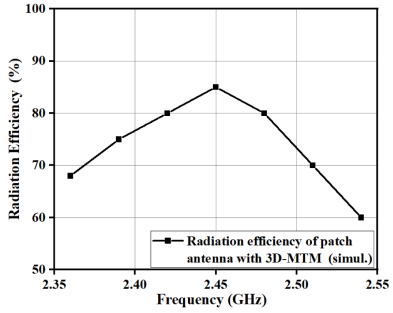


FIGURE 11. Simulated radiation efficiency of patch antenna with 3D-MTM superstrate.

In addition, by mounting the superstrate at an appropriate height from the patch antenna, the near-field effects are avoided. However, the microstrip patch antenna may become unstable due to mechanical instability, which will result in variations in resonant frequency, impedance mismatch, and affect the radiation pattern. To overcome these issues, it is necessary to select the appropriate substrate material as well as reinforce the support structure. Here, we have used the Roger 5880 substrate as a superstrate layer, which is well known for its mechanical rigidity and stability. Moreover, the foam support acts as an insulation layer, mitigating the impact of any mechanical vibrations on the microstrip patch antenna. There is a vibration-damping quality to the foam material used, which ensures that the mechanical energy is effectively absorbed and dissipated. So that any adverse effects on the patch antenna performances are minimized.

Figure 7(a) shows the measurement setup of the patch antenna with 3D-MTM superstrate. The S-parameters and radiation patterns are measured using a vector network analyzer and an anechoic chamber, available at the Antenna Design LAB, VIT, Vellore, India. Figure 7(b) shows measured S-parameters (S_{11}) of a patch antenna without and with a 3D-MTM superstrate. It is observed that the patch antenna and patch antenna with 3D-MTM superstrate are −32dB and −18dB, respectively at the resonant frequency of 2.45 GHz.

To better understand the influence of the 3D-MTM superstrate layer above the patch antenna for improvement of antenna gain. The electric field distribution of a patch antenna without and with a 3D-MTM superstrate is shown in Figure 8. A symmetric field distribution is observed in the patch antenna as shown in Figure 8(a). It can be seen from the analysis that, when the patch antenna is excited the surface current is distributed mainly on the center of the patch antenna. Figure 8(b) shows the maximum radiation occurring from the side of the patch antenna, and fields concentrated at the patch center are altered, which gives rise to a large amount of electric field distribution in the 3D-MTM superstrate layer. The fields coupled to each unit cell is traversed in opposite direction in one half of the 3D-MTM unit cell structure, as well as in adjacent unit cell which nullifies the effect of near-field coupling. Placing the superstrate structure above the patch antenna concentrates the coupled field primarily on the superstrate's unit cell surfaces, thereby increasing coupling fields and significantly enhancing the maximum radiation in the patch antenna. As a result, the gain and directivity of the antenna are increased. Figure 9 displays the simulated and measured radiation patterns in the E-plane for the patch antenna, both without and with the 3D-MTM superstrate. It shows that co-pol. radiation of the patch antenna without and with 3D-MTM superstrate are 3.4dBi and 8.4dBi, respectively. The simulated and measured results are matched in the patch antenna and patch antenna with 3D-MTM. The cross-pol. radiation of a patch antenna and patch antenna with 3D-MTM superstrate are about −30dB and −25dB, respectively. The half power beamwidth (HPBW) is decreased by about 23°. Figure 10. shows the gain comparison patch antenna without and with 3D-MTM superstrate. It is observed that the simulated and measured gain of the patch antenna is 3.4dBi and 3dBi, respectively at 2.45 GHz. The simulated and measured gain

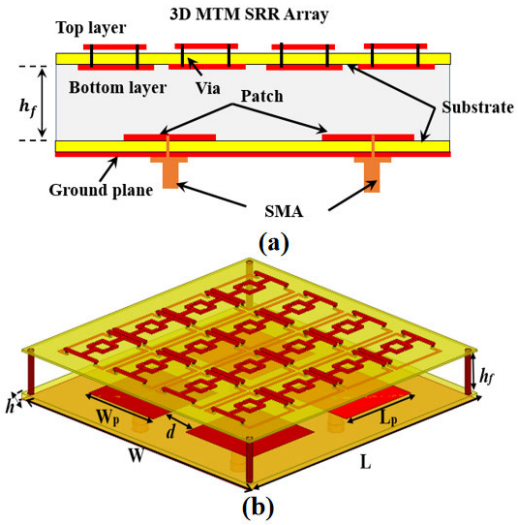


FIGURE 12. 2 × 2 array antenna (a) side view (b) 3D-view.

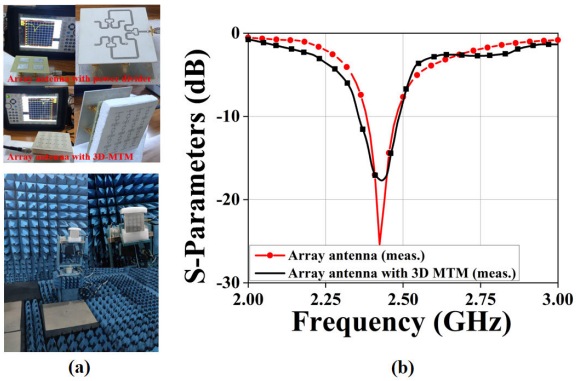


FIGURE 13. 2 × 2 array antenna (a) measurement setup with radiation chamber inset (b) measured S-parameters with 3D-MTM.

of patch antenna with 3D-MTM is about 8.4dBi and 8dBi, respectively at the resonant frequency of 2.45 GHz. The patch antenna gain is enhanced by about 5dBi. Figure 11. shows the simulated radiation efficiency of the patch antenna with 3D-MTM is about 85.

B. 2 × 2 ARRAY ANTENNA WITH 3D-MTM SUPERSTRATE

The illustration in Figure 12 displays the arrangement of a 2 × 2 array antenna with optimized dimensions, which incorporates with 3D-MTM superstrate. This antenna is designed and examined within the ISM band, specifically at a frequency of 2.45GHz. It utilizes an FR-4 substrate with a relative permittivity (ε_r) of 4.4 and a loss tangent (tan δ) of 0.02, and the substrate has a thickness of 1.6*mm*. The dimensions of the radiating elements and the ground plane are identical to those of a single-element patch antenna. The radiating elements and the ground plane of the array antenna are imprinted on both the upper and lower sides of the substrate. The separation distance (d) between the patch elements is set at $0.5\lambda_0$. To feed the 2 × 2 array antenna, a 1:4

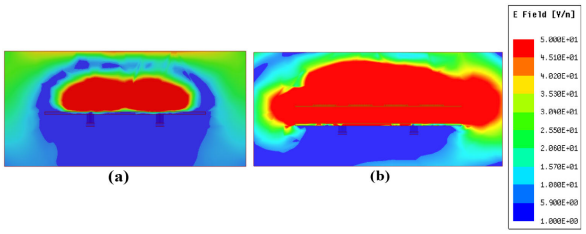


FIGURE 14. Electric field distribution (a) 2 ×2 array antenna 3D-MTM (b) 2 × 2 array antenna with 3D-MTM superstrate.

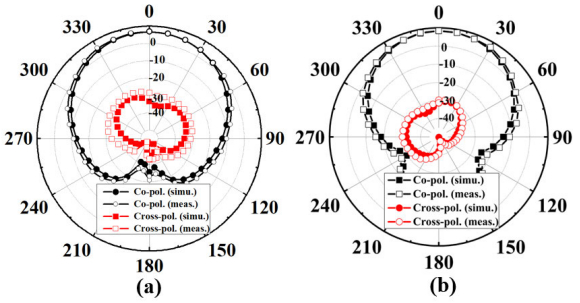


FIGURE 15. Simulated and measured E-plane radiation pattern (a) 2 ×2 array antenna (b) 2 × 2 array antenna with 3D-MTM superstrate.

Wilkinson equal power divider is employed. Additionally, to ensure impedance matching for the power dividers, λ/4 transformers are incorporated. The power divider has wide bandwidth from 1.5 to 3GHz with insertion loss of −6dB and isolation of −30dB. The fabricated prototype of a 2 × 2 array antenna with 3D-MTM superstrate is shown in Figure 13(a). The prototype is validated by measuring S-parameters and radiation patterns using a VNA and an anechoic chamber, respectively. The antenna is positioned inside an anechoic chamber with the support of wooden and foam structures, and it is connected through a 50Ω SMA connector. The measured S-parameters of the proposed 2 × 2 array antenna without and with 3D-MTM superstrate are shown in Figure 13(b). It shows a slight shift in resonance frequency of the 2 × 2 array antenna, due to the loading effect of 3D-MTM superstrate, fabrication tolerance, SMA connector soldering, or any outside influence on the input impedance. It is observed that the reflection coefficient of 2 × 2 array antenna without and with 3D-MTM superstrate is −24dB at 2.45GHz and −17dB at 2.43GHz. The electric field distribution of a 2 × 2 array antenna without and with 3D-MTM superstrate at the resonance frequency of 2.45GHz is presented in Figure 12. It can be observed that 2 × 2 array antenna has a uniform electric field distribution as shown in Figure 14(a), whereas in the case of with 3D-MTM superstrate loaded 2 × 2 array antenna the electric field is concentrated to the center of 3D-MTM superstrate surface. This is due to the uniform illumination effect of a 4 × 4 high negative refractive index superstrate and all the fields are coupled with the radiating patch so there is an increase in radiation as shown in Figure 14(b), which results in the high gain of an antenna.

TABLE 3. Performance comparison of proposed work with different existing MTM superstrate structure.

Ref	Freq (GHz) /BW (%)	MTM shape /Type	MTM characteristic	MTM array	No of superstrate	Antenna shape /Element	Antenna dimension	Enhanced Gain (dBi)
[13]	2.2 /2.7	Circular /2D	ZIM	9×9	2	Rectangle /Single	$2.68\lambda_o \times 2.68\ \lambda_o \times 0.71\lambda_o$	10
[15]	2.4 /3.3	Phi shaped /2D	DNG	5×5	1	Rectangular /Single	$0.48\lambda_o \times 0.48\ \lambda_o \times 0.013\lambda_o$	1.15
[19]	2.45 /2	Clover shaped /2D	ZIM	5×5	1	Rectangular /Single	$0.130\lambda_o \times 0.130\ \lambda_o \times 0.121\lambda_o$	6.12
[22]	2.4 &3.5 /2.9&8.5	Nested schurig/2D	ZIM	8×8	3	Monopole /Single	$1.2\lambda_o \times 1.7\ \lambda_o \times 1.1\lambda_o$	2.5 &3.2
[24]	2.45 /4.8	Nested SRR/2D	DPS	5×5	1	Circular /Single	$0.492\lambda_o \times 0.492\ \lambda_o \times 0.09\lambda_o$	5
[25]	9.4 /2	SRR /2D	NRI	7×7	4	Square /Single	$1.9\lambda_o \times 0.9\ \lambda_o \times 1.3\lambda_o$	10
[26]	6.18,9.65 /2.4,8.2	I-shape/2D	NZI	7×7	1	Rectangular /Single	$1.4\lambda_o \times 1.4\ \lambda_o \times 0.15\lambda_o$	4.1 &2.4
[27]	5.5 /6.9	Metal patch/2D	NRI	11×11	3	Square /Single	$1.9\lambda_o \times 1.9\ \lambda_o \times 1.01\lambda_o$	4
Proposed work	2.45 /5	3D-MTM SRR/3D	NRI	4×4	1	Square /Single	$0.82\lambda_o \times 0.82\ \lambda_o \times 0.11\lambda_o$	5
	2.45 /7	3D-MTM SRR/3D	NRI	4×4	1	Square / 2 × 2	$0.82\lambda_o \times 0.82 \times 0.11\lambda_o$	3.2

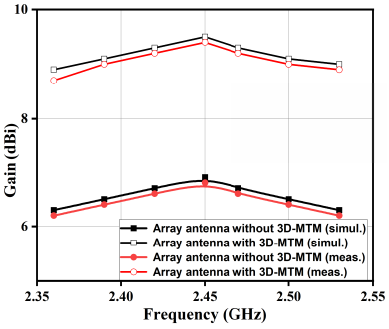


FIGURE 16. Gain comparison of 2 × 2 array antenna without and patch antenna with 3D-MTM superstrate.

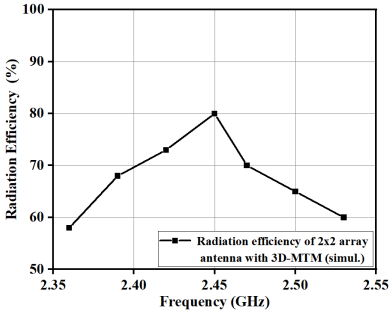


FIGURE 17. Simulated radiation efficiency of 2 × 2 array antenna with 3D-MTM superstrate.

The simulated and measured E-plane radiation patterns of 2 × 2 array antenna without and with 3D-MTM superstrate are shown in Figure 15. The co-pol. the radiation pattern of without and with 3D-MTM is 6.5dBi and 9.7dBi. The cross-pol. the radiation pattern is −35dB and -30dB. It is observed that array antenna gain is increased by about 3.2dBi and HPBW is decreased by about 12.85°. The 2D-gain comparison graph is shown in Figure 16. The simulated and measured gain of 2 × 2 array antenna without 3D-MTM is about 6.5dBi and 6.3dBi, and with 3D-MTM is about 9.7dBi and 9.5dBi, respectively at 2.45 GHz. The gain is enhanced by about 3.2dBi. Figure 17. shows the simulated radiation efficiency of 2 × 2 array antenna with 3D-MTM is about 80%. Table 3 compares the proposed and existing works.

V. CONCLUSION

A gain enhancement in a patch antenna and 2 × 2 array antenna with high refractive index 3D-MTM superstrate. This 3D-MTM exhibits a high refractive index within the frequency range of 2.36-2.53 GHz, making it effective as an insulating material to reflect surface waves. Therefore, the 4 × 4 3D-MTM array of the proposed unit cell is analyzed and employed as a superstrate for the patch antenna

and 2 × 2 array antenna resonant around 2.45 GHz. The patch antenna gain is enhanced by 5dBi and the reduced half power beam width in E-plane is 23°. The 2 × 2 array antenna gain is enhanced by about 3.2dBi and the half power beam width is 12.85°. This configuration proves to be a cost-effective, high-gain microstrip patch antenna suitable for 2.45 GHz WLAN applications, and it also exhibits enhanced suitability for modern wireless communication systems.

REFERENCES

[1] Y. Gong, S. Yang, B. Li, Y. Chen, F. Tong, and C. Yu, "Multi-band and high gain antenna using AMC ground characterized with four zero-phases of reflection coefficient," *IEEE Access*, vol. 8, pp. 171457–171468, 2020.

[2] T. Wu, J. Chen, and M.-J. Wang, "Multi-state circularly polarized antenna based on the polarization conversion metasurface with gain enhancement," *IEEE Access*, vol. 8, pp. 84660–84666, 2020.

[3] D. Chen, W. Yang, and W. Che, "High-gain patch antenna based on cylindrically projected EBG planes," *IEEE Antennas Wireless Propag. Lett.*, vol. 17, no. 12, pp. 2374–2378, Dec. 2018.

[4] A. S. Mekki, M. N. Hamidon, A. Ismail, and A. R. H. Alhawari, "Gain enhancement of a microstrip patch antenna using a reflecting layer," *Int. J. Antennas Propag.*, vol. 2015, pp. 1–7, Oct. 2015.

[5] Y. Yuan, X. Xi, and Y. Zhao, "Compact UWB FSS reflector for antenna gain enhancement," *IET Microw., Antennas Propag.*, vol. 13, no. 10, pp. 1749–1755, Aug. 2019.

[6] M. Zada, I. A. Shah, and H. Yoo, "Metamaterial-loaded compact high-gain dual-band circularly polarized implantable antenna system for multiple biomedical applications," *IEEE Trans. Antennas Propag.*, vol. 68, no. 2, pp. 1140–1144, Feb. 2020.

[7] W. Wu, B. Yuan, B. Guan, and T. Xiang, "A bandwidth enhancement for metamaterial microstrip antenna," *Microw. Opt. Technol. Lett.*, vol. 59, no. 12, pp. 3076–3082, Dec. 2017.

[8] T. Cai, G.-M. Wang, X.-F. Zhang, Y.-W. Wang, B.-F. Zong, and H.-X. Xu, "Compact microstrip antenna with enhanced bandwidth by loading magneto-electro-dielectric planar waveguided metamaterials," *IEEE Trans. Antennas Propag.*, vol. 63, no. 5, pp. 2306–2311, May 2015.

[9] D. Li, Z. Szabo, X. Qing, E.-P. Li, and Z. N. Chen, "A high gain antenna with an optimized metamaterial inspired superstrate," *IEEE Trans. Antennas Propag.*, vol. 60, no. 12, pp. 6018–6023, Dec. 2012.

[10] H. Zhou, Z. Pei, S. Qu, S. Zhang, J. Wang, Z. Duan, H. Ma, and Z. Xu, "A novel high-directivity microstrip patch antenna based on zero-index metamaterial," *IEEE Antennas Wireless Propag. Lett.*, vol. 8, pp. 538–541, 2009.

[11] T. Shaw, D. Bhattacharjee, and D. Mitra, "Gain enhancement of slot antenna using zero-index metamaterial superstrate," *Int. J. RF Microw. Comput.-Aided Eng.*, vol. 27, no. 4, Dec. 2016, Art. no. e21078.

[12] R. W. Ziolkowski and A. D. Kipple, "Application of double negative materials to increase the power radiated by electrically small antennas," *IEEE Trans. Antennas Propag.*, vol. 51, no. 10, pp. 2626–2640, Oct. 2003.

[13] Z. Weng, N. Wang, Y. Jiao, and F. Zhang, "A directive patch antenna with metamaterial structure," *Microw. Opt. Technol. Lett.*, vol. 49, no. 2, pp. 456–459, Feb. 2007.

[14] A. Bakhtiari, R. A. Sadeghzadeh, and M. N. Moghadasi, "Gain enhanced miniaturized microstrip patch antenna using metamaterial superstrates," *IETE J. Res.*, vol. 65, no. 5, pp. 635–640, Sep. 2019.

[15] M. Saravanan, V. B. Geo, and S. M. Umarani, "Gain enhancement of patch antenna integrated with metamaterial inspired superstrate," *J. Electr. Syst. Inf. Technol.*, vol. 5, no. 3, pp. 263–270, Dec. 2018.

[16] A. S. Dixit and S. Kumar, "Gain enhancement of antipodal Vivaldi antenna for 5G applications using metamaterial," *Wireless Pers. Commun.*, vol. 121, no. 4, pp. 2667–2679, Dec. 2021.

[17] H. Kavousi, J. Rashed-Mohassel, and M. Edalatipour, "Miniaturized patch antenna using a circular spiral-based metamaterial: KAVOUSI et al.," *Microw. Opt. Technol. Lett.*, vol. 59, no. 9, pp. 2276–2279, Sep. 2017.

[18] S. S. Al-Bawri, M. S. Islam, H. Y. Wong, M. F. Jamlos, A. Narbudowicz, M. Jusoh, T. Sabapathy, and M. T. Islam, "Metamaterial cell-based superstrate towards bandwidth and gain enhancement of quad-band CPW-fed antenna for wireless applications," *Sensors*, vol. 20, no. 2, p. 457, Jan. 2020.

[19] W. Lee, S.-I. Choi, H.-I. Kim, S. Hwang, S. Jeon, and Y.-K. Yoon, "Metamaterial-integrated high-gain rectenna for RF sensing and energy harvesting applications," *Sensors*, vol. 21, no. 19, p. 6580, Oct. 2021.

[20] K. Aydin, I. Bulu, and E. Ozbay, "Focusing of electromagnetic waves by a left-handed metamaterial flat lens," *Opt. Exp.*, vol. 13, no. 22, pp. 8753–8759, Oct. 2005.

[21] J. B. Pendry, "Negative refraction makes a perfect lens," *Phys. Rev. Lett.*, vol. 85, no. 18, pp. 3966–3969, Oct. 2000.

[22] Z. Haider, M. U. Khan, and H. M. Cheema, "A dual-band zero-index metamaterial superstrate for concurrent antenna gain enhancement at 2.4 and 3.5 GHz," *IETE J. Res.*, vol. 68, no. 4, pp. 2898–2908, Jul. 2022.

[23] S. Roy and U. Chakraborty, "Gain enhancement of a dual-band WLAN microstrip antenna loaded with diagonal pattern metamaterials," *IET Commun.*, vol. 12, no. 12, pp. 1448–1453, Jul. 2018.

[24] A. El Yassini, L. Aguni, S. Ibnyaich, S. Chabaa, and A. Zeroual, "Design of circular patch antenna with a high gain by using a novel artificial planar dual-layer metamaterial superstrate," *Int. J. RF Microw. Comput.-Aided Eng.*, vol. 29, no. 11, p. e21939, Nov. 2019.

[25] O. Borazjani, M. Naser-Moghadasi, J. Rashed-Mohassel, and R. Sadeghzadeh, "Design and fabrication of a new high gain multilayer negative refractive index metamaterial antenna for X-band applications," *Int. J. RF Microw. Comput.-Aided Eng.*, vol. 30, no. 9, p. e22284, May 2020.

[26] B. Ajewole, P. Kumar, and T. Afullo, "A microstrip antenna using I-shaped metamaterial superstrate with enhanced gain for multiband wireless systems," *Micromachines*, vol. 14, no. 2, p. 412, Feb. 2023.

[27] X. Gao, Y. Zhang, and S. Li, "High refractive index metamaterial superstrate for microstrip patch antenna performance improvement," *Frontiers Phys.*, vol. 8, p. 406, Sep. 2020.

[28] G. Al-Duhni and N. Wongkasem, "Metamaterial-inspired quintuple band printed patch antenna for dense communication networks," *J. Electromagn. Waves Appl.*, vol. 36, no. 18, pp. 2785–2803, Dec. 2022.

[29] G. H. Khouser and Y. K. Choukiker, "Design of negative refractive index 3D metamaterial split ring resonator for IoT applications," *Microw. Opt. Technol. Lett.*, vol. 66, no. 1, pp. 1–9, Aug. 2022.

[30] G. H. Khouser and Y. K. Choukiker, "Unique 3D metamaterial split ring resonator for wireless communication with high negative refractive index and for medium ratio," *Waves Random Complex Media*, pp. 1–15, Sep. 2022.

[31] C. A. Balanis, "Microstrip patch," in *Antenna Theory Analysis and Design*, 3rd ed. Hoboken, NJ, USA: Wiley, ch. 14, sec. 14.2, pp. 816–819.



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